CATHERINE WOLFRAM

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EMPLOYMENT

Yale University, New Haven, CT — July 2024 - Present
NSF Postdoctoral Fellow and Gibbs Assistant Professor — (on leave July 2025-July 2026)

ETH Zurich, Zurich, Switzerland — July 2025-July 2026
Postdoc

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA — August 2019 - May 2024
PhD in Mathematics
Advisor: Scott Sheffield
Thesis title: *Random geometry in two and three dimensions*

University of Chicago, Chicago, IL — September 2015 - June 2019
B.A. in Mathematics with Honors
Dean's list, 2015-2019

Stanford University Online High School — September 2011 - June 2015

ARTICLES

The Brownian bubble tea, with Gregory F. Lawler, Fredrik Viklund, and Yilin Wang. ArXiv preprint 2610.17990.

The multinomial dimer model, with Richard Kenyon. ArXiv preprint 2506.12171.

Epstein curves and holography of the Schwarzian action, with Franco Vargas Pallete and Yilin Wang. ArXiv preprint 2503.05690.

Large deviations for the 3D dimer model, with Nishant Chandgotia and Scott Sheffield. *Accepted* (to appear) in **Astérisque**.

Circle homeomorphisms with square summable diamond shears, with Dragomir Šarić and Yilin Wang. **Int. Math. Res. Not.** (IMRN) Vol. 2024, 17 (2024).

AWARDS

NSF Postdoctoral Fellowship — Fall 2024-Spring 2028

Baddoo Community Building Award — May 2024
MIT math department award

Phyllis Ruby Block Fellowship — Spring 2024
MIT math department fellowship

TEACHING

Yale:

- Instructor, Discrete mathematics (Math 244) — Fall 2026
- Instructor, Vector analysis and integration on manifolds (Math 302), — Spring 2027

MIT:

- Recitation instructor, probability and random variables (18.600) Fall 2023
  Instructor: Scott Sheffield
- Recitation instructor, multivariable calculus (18.02A) January 2023
  Instructor: Bill Minicozzi
- Recitation instructor, probability and random variables (18.600) Spring 2022
  Instructor: Scott Sheffield
- Recitation instructor, probability and random variables (18.600) Fall 2021
  Instructor: Jonathan Kelner

University of Chicago:

- Grader for Honors Calculus (Math 162) January 2019 - March 2019
- Course assistant (VCA) for Calculus (Math 151-152-153) September 2016 - June 2017

MENTORSHIP AND PROFESSIONAL EXPERIENCE

Referee for journal articles Periodic
Annals of Probability, Transactions of the American Mathematical Society, Discrete Mathematics, Proceedings of the London Mathematical Society

Defense committee member for graduate students
Raphael de Marreiros (PhD February 2025)

Mentor for students
Sumukh Ramakrishnan and Jiakai Li (ETH masters students, semester project, Spring 2026), June Lee (Yale undergraduate, reading project, summer 2025), MIT undergraduates through the directed reading program (2020-2024).

Mentor for Research Science Institute (RSI) hosted by MIT (summer 2020)

SERVICE AND ORGANIZATION

Co-organizer of the Friday Morning Seminar at Yale September 2024 - Present
Informal weekly seminar with some invited speakers and some informal discussions on topics in geometry, probability, combinatorics, dynamics, etc.

Organizer of informal seminars at Yale and ETH Spring 2025- Spring 2026
Started and ran an informal seminar at Yale (Spring 2025) and then at ETH (Fall 2025-Spring 2026), which met every week, for PhD students and postdocs to present on papers or their work in a range of topics (random conformal geometry, statistical mechanics, hyperbolic geometry, geometric analysis, etc...)

Member of the Yale math department climate committee September 2024 - May 2025
Department committee for organizing community events and outreach in the Yale math department.

Co-organizer and mentor, MIT Directed Reading Program (DRP) October 2022 - May 2024 Evaluated applications, paired undergraduate students with mentors, and arranged events. The program has about 100 students per year.

Co-organizer of PuMaGraSS at MIT September 2020 - May 2021
Weekly, casual seminar for MIT math graduate students to give talks to their peers, name stands for "Pure Math Graduate Student Seminar."

Manager of the Math Learning Center (MLC) at MIT August 2020 - May 2021
Ran the math department's peer-tutoring center, managing 10-12 undergraduate tutors.

TALKS

Past and upcoming talks as of now, listed in reverse chronological order.

1. Yale undergraduate math society (expository talk for undergraduates) — October 2026
2. Cambridge equinox probability conference, Cambridge, UK — September 2026
3. One World Probability Seminar (online talk) — April 2026
4. Random Geometry in Math and Physics, Simons Center, Stony Brook, NY — April 2026
5. EPFL probability seminar, Lausanne, Switzerland — March 2026
6. University of Bern IMSV colloquium, Bern, Switzerland — March 2026
7. Les Diablerets workshop on mathematical physics, Les Diablerets, Switzerland — February 2026
8. IPAM workshop: New interactions between probability and geometry, Los Angeles, CA — January 2026
9. KTH probability and mathematical physics seminar, Stockholm, Sweden — December 2025
10. Vienna probability seminar, Vienna, Austria — November 2025
11. Universal Teichmüller theory workshop, Aussois, France — August 2025
12. CIRM perfectly matched perspectives in statistical mechanics, combinatorics and geometry, Marseille, France — June 2025
13. AMS sectional meeting session on Probability and Combinatorics Hartford, CT — April 2025
14. University of Michigan probability seminar, Ann Arbor, MI — February 2025
15. Princeton probability seminar, Princeton, NJ — November 2024
16. Columbia probability seminar, New York, NY — October 2024
17. AdS geometry reading group (two expository talks, online) — October 2024
18. Northwestern probability seminar, Evanston, IL — September 2024
19. 11th World Congress on Probability and Statistics, Bochum, Germany — August 2024
20. Workshop on Probabilistic Field Theories, Aalto University — June 2024
21. Statistical physics and random surfaces, Oberwolfach — May 2024
22. IPAM workshop: Statistical Mechanics Beyond 2D, Los Angeles, CA — May 2024
23. Seed seminar of Mathematics and Physics (online) — April 2024
24. PuMaGraSS seminar at MIT (expository talk) — March 2024
    SLE curves, large deviations, and geometry
25. KPZ meets KPZ workshop, Fields Institute, Toronto, Canada — March 2024
26. University of Edinburgh probability seminar (online) — January 2024
27. MIT probability seminar, Cambridge, MA — November 2023
28. University of Chicago probability and statistical physics seminar Chicago, IL — November 2023
29. Brown combinatorics seminar, Providence, RI — November 2023

30. Informal seminar: dynamics, geometry, and moduli spaces at Harvard — November 2023
31. ICERM workshop: Asymptotic Limits of Discrete Random Structures — September 2023
    Providence, RI
32. Harvard CMSA probability seminar, Cambridge, MA — September 2023
33. MIT integrable probability seminar, Cambridge, MA — September 2023
34. dimers2023 conference, Sorbonne Universite, Paris, France — July 2023
35. Percolation today (online) — June 2023
36. LA probability forum (joint between Caltech, UCLA, and USC) — May 2023
37. IHES informal analysis and probability seminar, Bures sur Yvette, France — April 2023
38. Courant probability and mathematical physics seminar, New York, NY — March 2023
39. Rutgers geometry/topology seminar (online) — March 2023
40. Informal seminar: dynamics, geometry, and moduli spaces at Harvard — February 2023
41. CUNY Hyperbolic geometry seminar, New York, NY — February 2023
42. Random Geometry and Statistical Physics Workshop, University of Pennsylvania — October 2022
43. Probability and Mathematical Physics conference in Helsinki, Finland — July 2022
44. PuMaGraSS seminar at MIT (expository talk) — April 2022
    Gluing polygons, Wick's formula, and moduli spaces
45. Reading group on the dimer model at MIT (expository talk) — March 2022
    Kasteleyn theory on the torus
46. PuMaGraSS seminar at MIT (expository talk) — April 2021
    The dimer model in 2D and 3D, local moves, and topology
47. AWM Women in Math Symposium, University of Chicago — February 2017
    Recursive derivation of 2n-gon topologies.

ACADEMIC VISITS

Institute for Advanced Study, Princeton, NJ — September 2022 - December 2022
Visiting graduate student while my advisor was there as a visiting professor.

Institut des Hautes Études Scientifiques, Bures sur Yvette, France — April - May 2023; June 2024
Invited researcher visiting Prof. Yilin Wang.

CONFERENCES AND SUMMER SCHOOLS

A more complete list of conferences, summer schools, and workshops I've attended or plan to attend as of now, in reverse chronological order.

Statistical physics and random surfaces, Oberwolfach, Germany — May 2027

Quantum Fields, Probability, and Geometry program, Institut Mittag Leffler, Sweden — March 2027

Cambridge equinox probability conference, Cambridge, UK — September 2026

Random Explorations: From Random Walks to Random Geometry, IMSI, Chicago, IL — July 2026

Random Geometry in Math and Physics, Simons Center, Stony Brook, NY — April 2026

Les Diablerets workshop on mathematical physics, Les Diablerets, Switzerland February 2026

IPAM workshop: New interactions between probability and geometry, Los Angeles, CA January 2026

Universal Teichmüller theory workshop, Aussois, France August 2025

CIRM perfectly matched perspectives on statistical mechanics, combinatorics and geometry June 2025

AMS sectional meeting session on Probability and Combinatorics April 2025

Complex Faces, Venice, Italy April 2025

Northeast Probability Seminar, New York, NY November 2024

11th World Congress on Probability and Statistics, Bochum, Germany August 2024

Workshop on Probabilistic Field Theories, Aalto University, Espoo, Finland June 2024

Statistical physics and random surfaces, Oberwolfach, Germany May 2024

Statistical mechanics beyond 2D, IPAM, Los Angeles, CA May 2024

KPZ meets KPZ workshop, Fields Institute, Toronto, Canada March 2024

Probability and geometry in, on and of non-Euclidean spaces, CIRM, Marseille, France October 2023

Asymptotic limits of discrete random structures, ICERM, Providence, RI September 2023

dimers2023 conference, Sorbonne Universite, Paris, France July 2023

Three facets of gravity workshop, Humboldt University, Berlin, Germany May 2023

Random Geometry and Statistical Physics Workshop, University of Pennsylvania, PA October 2022

Cornell Probability Summer School (as a TA), Cornell University, Ithaca, NY August 2022

Probability and Mathematical Physics conference, Helsinki, Finland July 2022

The Analysis and Geometry of Random Spaces, MSRI, Berkeley, CA March 2022
Program associate for two weeks of the semester program (including the one week workshop)

Random Geometry conference, CIRM, Marseille, France January 2022

Random Conformal Geometry Summer School, MSRI, Berkeley, CA (virtual) July 2021

OTHER

Citizenships: United States, United Kingdom